EMBEDDING AN ABELIAN CATEGORY INTO THE ABELIAN SHEAVES ON ITS ASSOCIATED SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the exactness and full faithfulness, the only delegations being named results carried on this card as navigation stubs; the closing sentence about enough points names the criterion it applies and is preserved verbatim.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The site attached to an abelian category. Same chapter, Lemma lemma-site-abelian-category. Supplies the site structure used throughout, whose coverings make the representable presheaves sheaves.
- Fibre products and the exactness criterion. Homology, Example homology-example-fibre-product-pushouts and Lemma homology-lemma-check-exactness. Supply the description of the fibre product of a surjection as a kernel and the criterion turning exactness into exactness of the induced sequences of Hom groups.
- The universal property of epimorphisms. Homology, Lemma homology-lemma-epimorphism-universal-abelian-category. Supplies that the projection from the fibre product along a surjection is again a surjection, which produces the covering used for right exactness.
- Monomorphisms and epimorphisms of sheaves. Sites, Lemma sites-lemma-mono-epi-sheaves and Proposition sites-proposition-criterion-points. Supply the characterization of surjectivity of a map of sheaves by local lifting and the criterion for a site to have enough points.
- The Yoneda lemma. Categories, Lemma categories-lemma-yoneda. Supplies the full faithfulness of the Yoneda embedding used in the proof.

Date: stacks-project:injectives.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05PM: sources/stacks-injectives/pinned-original-source.tex, source lines 1267–1267. Navigation pointer invoked at proof line 1296; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 05PJ: sources/stacks-homology/pinned-original-source.tex, source lines 918–918. Navigation pointer invoked at proof line 1308; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05AA: sources/stacks-homology/pinned-original-source.tex, source lines 967–967. Navigation pointer invoked at proof line 1328; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 001P: sources/stacks-categories/pinned-original-source.tex, source lines 758–758. Navigation pointer invoked at proof line 1334; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05PK: sources/stacks-homology/pinned-original-source.tex, source lines 1136–1136. Navigation pointer invoked at proof line 1283; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WN: sources/stacks-sites/pinned-original-source.tex, source lines 2473–2473. Navigation pointer invoked at proof line 1346; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original proposition 00YQ: sources/stacks-sites/pinned-original-source.tex, source lines 9618–9618. Navigation pointer invoked at proof line 1350; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C} = (\mathcal{A}, Cov)$ be the site defined in Lemma 05PM. Then $X \mapsto h_X$ defines a fully faithful, exact functor*

$$\mathcal{A} \longrightarrow Ab(\mathcal{C}).$$

Moreover, the site $\mathcal{C}$ has enough points.

Proof. Suppose that $f : V \to U$ is a surjective morphism of $\mathcal{A}$. Let $K = \mathrm{Ker}(f)$. Recall that $V \times_U V = \mathrm{Ker}((f, -f) : V \oplus V \to U)$, see Homology, Example 05PJ. In particular there exists an injection $K \oplus K \to V \times_U V$. Let $p, q : V \times_U V \to V$ be the two projection morphisms. Note that $p - q : V \times_U V \to V$ is a morphism such that $f \circ (p - q) = 0$. Hence $p - q$ factors through $K \to V$. Let us denote this morphism by $c : V \times_U V \to K$. And since the composition $K \oplus K \to V \times_U V \to K$ is surjective, we conclude that c is surjective. It follows that

$$V \times_U V \xrightarrow{p-q} V \to U \to 0$$

is an exact sequence of $\mathcal{A}$. Hence for an object X of $\mathcal{A}$ the sequence

$$0 \to \mathrm{Hom}_{\mathcal{A}}(U, X) \to \mathrm{Hom}_{\mathcal{A}}(V, X) \to \mathrm{Hom}_{\mathcal{A}}(V \times_U V, X)$$

is an exact sequence of abelian groups, see Homology, Lemma 05AA. This means that h_X satisfies the sheaf condition on $\mathcal{C}$.

The functor is fully faithful by Categories, Lemma 001P. The functor is a left exact functor between abelian categories by Homology, Lemma 05AA. To show that it is right exact, let $X \to Y$ be a surjective morphism of $\mathcal{A}$. Let U be an object of $\mathcal{A}$, and let $s \in h_Y(U) = \mathrm{Mor}_{\mathcal{A}}(U, Y)$ be a section of h_Y over U. By Homology, Lemma 05PK the projection $U \times_Y X \to U$ is surjective. Hence $\{V = U \times_Y X \to U\}$ is a covering of U such that $s|_V$ lifts to a section of h_X. This proves that $h_X \to h_Y$ is a surjection of abelian sheaves, see Sites, Lemma 00WN.

The site $\mathcal{C}$ has enough points by Sites, Proposition 00YQ. □